ZEERBOX · CONCEPT OPTIONS

ZeerBox concept options

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DOCUMENT CONTROL

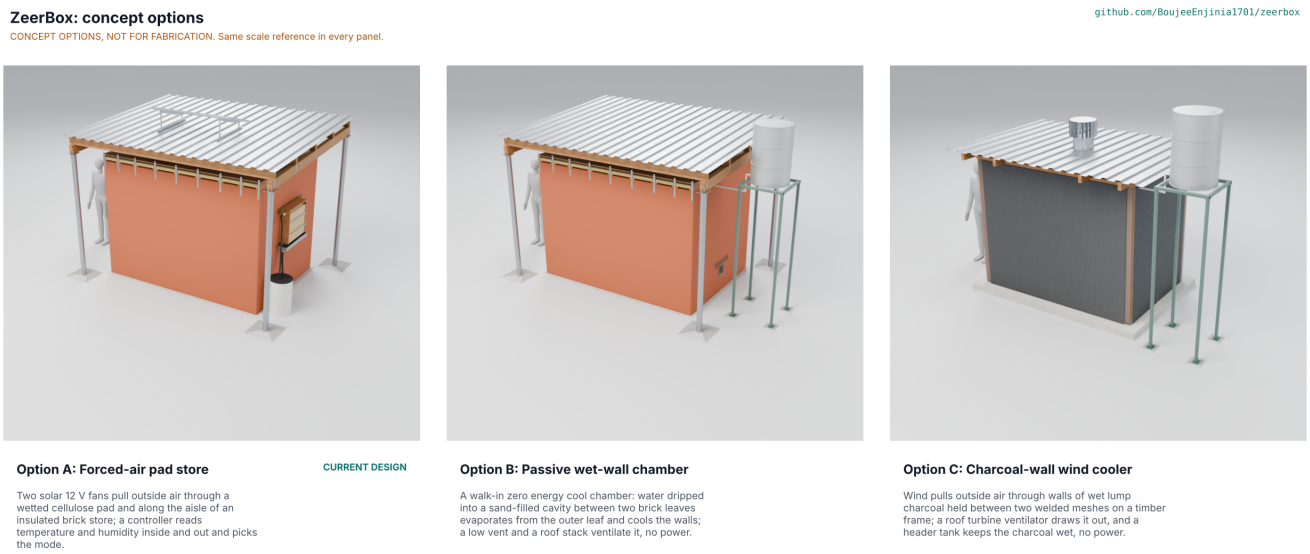
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REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-24	Amish Chadha	First issue (kit 1.9.1 upgrade): the existing design as option A, two options with different working principles, all three modelled at detail level D3, scored against ZBX-URS-001

ZeerBox concept options

Option A, the existing forced-air pad store, scores 190 of 235 against the user requirements; option B, a passive wet-wall chamber, and option C, a charcoal-wall wind cooler, both score 173 (8.9 % below A). All three are within 10 % of the top score, so under STANDARDS section 21 all three go to Amish without a pick (Proposed, awaiting Amish; ZBX-DDR-004). Option A is the current design and stays the design of record unless he decides otherwise; it is the only option that meets the two promises in the pitch, a cooling drop of 8 K or more (PR-002) and a store that tells the user when it is not cooling (PR-004). Method: STANDARDS section 21.



The options picture is rendered in Blender Cycles from `components()` *of every option in* `cad/src/model.py` *by* `python .kit/options.py`*, which runs on Amish's Mac; until then the image may be missing.*

All three options share the brief: about 24 crates (480 kg) of mixed vegetables in a walk-in room 2.4 x 1.5 x 2.0 m inside, the 35 °C, 30 % RH design day, 100 kg of warm produce loaded a day, the same timber racks and the same door. They differ in how they cool.

The options

Option A: forced-air pad store (the existing design)

Two solar 12 V fans pull outside air through a wetted cellulose pad and along the aisle of an insulated brick store; a controller reads temperature and humidity inside and out and picks the mode. Double fired-brick walls with a dry rice husk cavity, a framed and insulated ceiling, a shade roof on four posts carrying a 150 W panel, a 150 mm pad in the back wall fed from a 60 L sump, two 250 mm fans in a ceiling fan box at the door end, and a controller with a three-color mode lamp. On the design day the store sits at 24.3 °C, 10.7 K below the outside air (9.8 to 11.4 K), at 81 % RH, using 43 L of water and 372 Wh of solar energy (ZBX-CAL-001 v0.4). Key risks: assumed fan and pad curves (STM S3, S5), humidity at the low edge of its target (PR-003), a sealed room if the battery runs flat (ZBX-DEC-001 D3). From the partner: co-design of size, ownership and crops; a pad supplier; a site.

Option B: passive wet-wall chamber

A walk-in zero energy cool chamber: water dripped into a sand-filled cavity between two brick leaves evaporates from the outer leaf and cools the walls; a low vent and a roof stack ventilate it, with no power. It reuses option A's room, ceiling, roof, door and racks; the cavity holds wet sand, a perforated drip ring lies on the sand under the capping course, a 200 L drum on a 2.2 m welded stand feeds it by gravity, a 200 x 200 mm meshed vent low in the back wall lets air in and a 110 mm stack through the ceiling lets it out, and a dial thermometer and hygrometer hangs by the door. First-order balance (STM B1 to B3): a full store with the day's warm produce is held only about 3 K below outside air (ZECCs with lighter loads measure 4 to 7 K), at 55 to 90 % RH depending on how wet the inner leaf runs, with about 3 air changes an hour that drain cool air out of the low vent and draw 35 °C air in at the stack, using 60 to 70 L of water a day that must be lifted 2.2 m into the tank. Key risks: too little cooling for a full walk-in load; mold in a damp, slowly ventilated room (B5); daily water lifting. From the partner: whether growers accept less cooling for no electrics; watering routine.

Option C: charcoal-wall wind cooler

Wind pulls outside air through walls of wet lump charcoal held between two welded meshes on a timber frame; a roof turbine ventilator draws it out, and a header tank keeps the charcoal wet, with no power. A concrete slab with eight treated timber posts on galvanised post bases, 100 mm charcoal walls between inner and outer galvanised welded mesh, top plates, a drip pipe threaded through the posts along each wall top, a sheeted roof on four rafters with a 300 mm turbine ventilator, a framed door holding a charcoal panel, and a 200 L tank on a 2.25 m stand. First-order balance (STM C1 to C3, C1 conditional on the airflow): in a light wind about 7 K below outside air, using 30 to 40 L of water a day; on a calm afternoon the turbine barely turns and cooling falls away (the airflow is to assess, C2). Key risks: dependence on wind; galvanised mesh corroding in wet charcoal (C4); mold in the charcoal (C5); a light structure with a 10-year life in doubt. From the partner: local experience with charcoal coolers (they are used for milk and vegetables in Kenya); charcoal and mesh supply.

First-order numbers

Table 1. Options compared. Costs are estimates from the repo BOM's unit prices and cited benchmarks (ZBX-MKT-001), not quotes; masses and part counts are from `cad/fidelity.json`.

Quantity	Option A	Option B	Option C
Estimated parts cost (USD)	874 (kit 300, structure 574; ZBX-CAL-001 s11)	About 670 (option A's structure less the pad, fan box and kit, plus wet sand, drip ring, tank and stand, vent and stack)	About 520 (slab, timber frame, mesh, charcoal, roof, turbine ventilator to assess, tank and stand, racks)
Mass (kg)	12,480	15,041	2,468
Output: drop below outside air at the design point	10.7 K (9.8 to 11.4 K)	About 3 K (full load, first order)	About 7 K in a light wind; little in calm air
Store RH while cooling	81 % (77 to 85 %)	55 to 90 % (to assess, STM B4)	About 85 to 90 % (estimate)
Energy or effort	372 Wh a day from a 150 W panel; 43 L of water poured into a sump at 0.6 m	No power; 60 to 70 L of water a day lifted 2.2 m	No power; 30 to 40 L of water a day lifted 2.25 m
Parts count (D3 model)	1,181 parts (817 fasteners)	1,004 parts (706 fasteners)	538 parts (343 fasteners)
STEM check (ZBX-STM-001)	Within known science; 6 plausible rows need data; two fixes made (D1, D5)	Within known science; B4 and B5 plausible	Within known science; C1, C2, C4, C5 plausible (cooling rests on the ventilator's airflow)

Scoring

Table 2. Weighted scoring (weights: High UR 3, Medium 2, Low 1; scores 1 to 5).

Criterion	Source	Weight	A	B	C	Reason
Hold several days of harvest	UR-001	3	5	5	5	Same 24-crate room in all three
Load without heavy lifting	UR-002	2	5	5	5	Same racks, top shelf at 1.40 m
Noticeably cooler on hot dry days	UR-003	3	5	2	3	10.7 K against about 3 K (B, full load) and about 7 K in wind (C)
Produce does not wilt	UR-004	3	4	4	4	A at 81 % RH, at risk; B 55 to 90 % depending on how wet its inner leaf runs (STM B4); C high but wind dries it
Know when it is not cooling	UR-005	3	5	2	2	A's controller and mode lamp; B and C only a dial the user must read and interpret
No mold and safe air	UR-006	3	4	2	3	A has a guard mode and 8 air changes an hour in hold; B is damp and slowly ventilated; C is airy but the charcoal stays wet
Produce lasts longer	UR-007	3	4	3	3	Follows cooling and humidity; ZECC evidence is for lighter loads
Works without the grid	UR-008	3	4	5	5	A needs its solar kit and battery; B and C need no power
Water I can carry	UR-009	2	4	1	2	A 43 L poured into a ground-level sump; B 60 to 70 L and C 30 to 40 L lifted into a tank above head height every day
Safe to touch and work on	UR-010	3	4	5	5	A is 12 V with guarded fans; B and C have no electrics
Built by local trades	UR-011	3	4	5	4	A adds a 12 V installation; B is masonry and plumbing; C needs mesh work and a charcoal supply
Mended from the regional town	UR-012	3	3	4	3	A has fans, pump, sensors and pad to replace; B only a drip ring and tank; C mesh and charcoal
Lasts in heat, dust, termites and wind	UR-013	2	4	3	2	A's dry brick and steel roof; B's walls stay wet; C's timber and mesh sit in wet charcoal
Affordable to a group	UR-014	3	3	4	4	About USD 874, 670 and 520
Cost against the value-engineering target	project.yaml	2	3	5	5	A's kit is at the USD 300 target; B's and C's water kits are well under it
Buildable with local skills and tools		2	4	5	4	As UR-011, from the build side
Safety	ZBX-RSK-001	3	4	3	4	A: battery and fans managed, sealed-room risk when flat (D3); B: damp sealed room; C: open structure
Maintenance		1	3	3	2	A: weekly sump and pad routine; B: daily water lifting; C: daily water and charcoal renewal

Criterion	Source	Weight	A	B	C	Reason
Weighted total (of 235)			**190**	**173**	**173**	

Sensitivity: if the pitch's two promises (UR-003 cooling, UR-005 knowing the mode) are treated as must-meet, options B and C fail both and only option A remains. If the store must work with no electrical parts at all, option A drops out and option C leads B on cooling while B leads C on durability.

Recommendation

All three options are within 10 % of the top score, so all three go to Amish without a pick (STANDARDS section 21), recorded as "Proposed, awaiting Amish" in ZBX-DEC-001 (D2) and argued in ZBX-DDR-004. Option A remains the design of record: it is the current design, the only one that meets PR-002 and PR-004, and the repo is not redesigned without Amish's go.

The runner-up's best idea to graft onto option A, whichever is chosen: option B's passive ventilation path (a low vent and a stack), topped with option C's turbine ventilator head, as the always-open air path that keeps the room safe to enter when option A's battery is flat. That graft is the subject of ZBX-DEC-001 D3, proposed to Amish with this recommendation because it lets some cooled air out on design days and so is a safety trade-off.